Jimma University, Jimma Institute of Technology					
1	Faculty/school	Faculty of Computing and Informatics		Department	Software Engineering
2	**Course Information**				
	Course Name	Fundamentals of Machine Learning			
	Course Code	SEng4091			
	ECTS	5	Credit hours	3	
	Contact Hours	Lec	Lab	Tur	Total/Per Week
		2	3	0	5
	Year & Semester Offered:		Year: IV	**Semester:** II	
	Module Name			Status of the course:	
	Prerequisite:	SEng3092: Fundamentals of Artificial Intelligence			
3	**Instructor Information**				
	Instructor Name		Alemisa Endebu (MSc.)		
	Relevant Contact Information		alemisa.endebu@ju.edu.et		
	Office Location & Office No.		Admin Building,		
4	Course Description				
	• This course will introduce the field of Fundamentals of Machine Learning (ML), in particular focusing on the core concepts of supervised learning, unsupervised learning and Reinforcement Learning, Overview of Neural networks and Model evaluation. • The course cover different type of ML and different techniques of model evaluations. The practical will concern the application of ML to a range of real-world problems.				
5	**Course Learning Outcomes (CLOs):** At the end of the course the student will be able to attain:				
	CLO-1	• *Understand the mathematical foundations underlying ML algorithms and* its major learning paradigms.			
	CLO-2	• Implement appropriate supervised, unsupervised learning and Reinforcement learning algorithms to solve real-world problems.			
	CLO-3	• Evaluate and compare machine learning models using appropriate validation methods and performance metrics.			
	CLO-4	• Explain, design, implement, train, and evaluate deep learning models for solving real-world machine learning problems.			
	CLO-5	• Apply data preprocessing and feature engineering techniques to prepare datasets for machine learning.			
	CLO-6	• Develop and optimize machine learning models through algorithm selection, parameter tuning, and model evaluation.			
	CLO-7	• Analyze ethical, fairness, privacy, bias, and generalization issues in machine learning applications.			
6	Program Learning Outcomes (PLOs): Selected PLOs				
	PLO-2:	• An ability to identify, formulate, and solve complex engineering problems by applying principles of software engineering, science, and mathematics.			
	PLO-4:	• An ability to develop and conduct appropriate experimentation, analyses and interpret data, and use software engineering judgment to conclude.			

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	PLO-5:	• An ability to create, select, apply, and recognize limitations of appropriate techniques, resources, and modern engineering and IT tools, including prediction and modelling, to complex engineering problems.

7 **Mapping of the course Learning Outcomes(CLOs) to the Program Learning Outcomes(PLOs),** (Sample Mapping scale: strong alignment = ✓✓, moderate alignment = ✓)

CLOs	Program Learning Outcomes (PLOs)											Teaching Methods				Assessment Methods					
	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	L	Tut.	L/P	H	Tests	Mid-Exam	Lab exams	Final-Exam	Assignment	Project
CLO-1		✓										✓				✓	✓				
CLO-2		✓✓										✓	✓	✓		✓	✓	✓	✓	✓	✓
CLO-3		✓✓										✓	✓	✓		✓		✓	✓	✓	✓
CLO-4		✓✓										✓	✓	✓				✓	✓	✓	✓
CLO-5			✓✓	✓✓								✓	✓	✓				✓	✓	✓	✓
CLO-6				✓✓								✓	✓	✓				✓	✓	✓	✓
CLO-7					✓✓							✓	✓	✓							✓

8 **Transferable Practical (Skills if applicable)**

1.	Programming and coding skills
2.	Problem-solving skills
3.	Data analysis skills
4.	Model Evaluation skills
5.	Critical thinking
6.	Technical communication
7.	Research skills

9 **Distribution of Student Learning Time (SLT),** ***(allocate time for each teaching activities)***

Course Content Outline	Attainable CLOs	Teaching and learning methods				Assessment methods
		L	Tut	L/P	H	
Chapter 1: Introduction to Machine Learning 1.1 What is Machine learning? 1.2 History and relationships to other fields 1.3 Essential math and statistics for ML. 1.4 Applications of ML 1.5 Overview of types of ML techniques.	• **CLO-1**	2hrs				• Mid-Exam
Chapter 2: Supervised Learning 2.1 Introduction to Supervised Learning 2.2 Linear model *2.2.1. Regression* 2.2.2. Understand the operation of regression 2.2.2.1 Linear regression 2.2.2.2 Polynomial regression 2.2.2.3 Regularization techniques	• **CLO-2** • **CLO-3** • **CLO-6**	6hrs	2hrs	12hrs		• Mid-Exam • Assign • Lab-exam • Project

2.2.2.4 Understand the metrics used to evaluate regression 2.2.2.5 A case study in regression 2.2.3. *Classification* 2.2.4Understand the operation of classifiers 2.2.4.1 KNN 2.2.4.2 Naïve Bayes 2.2.4.3 Logistic regression 2.2.4.4 Decision trees 2.2.4.5 Random forest 2.2.4.6 Support vector machines 2.2.5A case study in classification 2.2.6Understand the metrics used to evaluate classifiers 2.3 How to improve supervised models 2.4 Parametric models for classification and regression 2.5 Understand the problems of over-parameterization and the curse of dimensionality 2.6 Use regularization on over-parameterized models						
Chapter 3: Unsupervised learning 3.1. Introduction to Unsupervised Learning 3.2. Understand the principles of unsupervised learning models 3.3. Clustering approaches 3.3.1. K-Means 3.3.2. K nearest neighbors 3.3.3. Hierarchical clustering 3.4. Correctly apply and evaluate clustering models 3.5. Association rule learning 3.5.1. Apriori algorithm	• **CLO-2** • **CLO-3** • **CLO-6**	5hrs		6hrs		• Assign • Lab-exam • Project • Final-Exam
Chapter 4: Reinforcement Learning 4.1. Introduction to Reinforcement learning 4.2. Terminologies Used in Reinforcement Learning. 4.3. Elements of Reinforcement Learning 4.4. Reinforcement learning *4.4.1. Markov decision* *4.4.2. Q-Learning* *4.4.3. Monte Carlo Prediction* 1.5. Applications of Reinforcement Learning 1.6. Case study	• **CLO-2** • **CLO-3** • **CLO-6**	4hrs		6hrs		• Assign • Lab-exam • Project • Final-Exam
Chapter 5: Neural Network 5.1. Introduction NN 5.1.1. Understanding the Brain 5.1.2. Neural networks as a paradigm for parallel processing 5.2. The Perceptron 5.3. Training a Perceptron 5.4. Artificial neural network 5.5. Multilayer Perceptron 5.6. Backpropagation algorithm 5.6.1. Nonlinear Regression 5.6.2. Two-Class Discrimination 5.6.3. Multiclass Discrimination 5.6.4. Multiple Hidden Layers 5.7. Recurrent and Sequence-Based Deep Learning Models 5.7.1. Recurrent Neural Networks (RNNs)	• **CLO-4** • **CLO-6**	6hrs	2hrs	12hrs		• Assign • Lab-exam • Project • Final-Exam

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	5.7.2. Long Short-Term Memory (LSTM) 5.7.3. Gated Recurrent Unit (GRU) 5.7.4. Sequence-to-Sequence (Seq2Seq) Models Reading assignment 5.7.5. Transformer-based models 5.8. Convolutional Neural Networks (CNNs) 5.8.1. What is CNN? 5.8.2. Why CNN? 5.9. Training procedures 5.7.1. Improving convergence 5.7.2. Overtraining 5.7.3. Structuring the network 5.10. Tuning the network size 5.11. A case study in neural networks						
	Chapter 6: Model Evaluation 6.1. Data processing 6.2. Data cleaning and transforming 6.3. Feature selection and visualization 6.4. Model selection and tuning 6.5. Methods of dimensional reduction *6.5.1. Principal Component Analysis (PCA)* *6.5.2. Singular value decomposition (SVD)* *6.5.3. T-distributed Stochastic Neighbor Embedding (t-SNE)* *6.5.4.* Optimize the performance of the model *6.5.5.* Control model complexity 6.6. Over-fitting and Under-fitting 6.7. Cross-Validation and Re-sampling methods *6.7.1.1. K-Fold Cross-Validation* *6.7.1.2. 5 ×2 Cross-Validation* *6.7.1.3. Bootstrapping* 6.8. Gradient descent (batch, stochastic) 6.8.1. Bias, variance 6.9. Performance Evaluation Methods 6.10. Tool kit	• **CLO-3** • **CLO-5** • **CLO-6** • **CLO-7**	6hrs	2hrs	6hrs		• Mid-Exam • Assign • Lab-exam • Project • Final-Exam

10	**Distribution of laboratory practice** ***(when applicable)***			
	Lab activities Outline		**Attainable CLOs**	
	Lab-1: Basic Python Programming		CLO-2	
	Lab-2: Data Annotation and Data Preprocessing		CLO-3, CLO-5, CLO-6, CLO-7	
	Lab-3: Supervised Machine Learning		CLO-2, CLO-6	
	Lab-4: Unsupervised Machine Learning		CLO-2, CLO-6	
	Lab-5: Reinforcement Learning		CLO-2, CLO-6	
	Lab-6: Neural Networks and Model Evaluation		CLO-4, CLO-6,CLO-7	
11	**Assessments**	**Percentage (100%)**	**F2F(Face-to-Face)**	**NF2F**(Non-Face-to-Face)
	Continues assessment	10%	✓	✓
	Mid Term Exam	20%	✓	
	Project	25%		✓
	Final Exam	45%	✓	
12	Reference and text book		**List of Text books:** 1. Shalev-Schwartz & Ben-David, Understanding Machine Learning. Cambridge University Press. 2014. 2. Hastie, Tibshirani & Friedman, The Elements of Statistical Learning: Data Mining, Inference, and Prediction, 2nd Edition, Springer, 2009	

		List of Reference books 1. Bishop, C. (2006), Pattern Recognition and Machine Learning, Berlin: Springer-Verlag 2. Aurélien Géron — Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow, 3rd Edition (2022/2023) 3. Deep Learning: An MIT Press book-Ian Goodfellow and Yoshua Bengio and Aaron Courville
	Key: L = Lecture, T = Tutorial, L/P = lab practice, H = home study, F2F = Face to Face, NF2F = Non Face to Face	

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